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CO3 Assessment Tool 3: Error Analysis Task

ECA0905 – Digital Signal Processing

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Question

- A Burg spectral estimation algorithm uses model order 2 instead of 10, producing poor spectral peaks. Identify the implementation error.



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Implementation Error

- **Primary Issue:** The Burg algorithm was executed with an Autoregressive (AR) order of $p = 2$ instead of $p = 10$.
- **Type of Error:** Model Under-parameterization / Underfitting
- **Direct Impact on Spectrum:**
 - Insufficient degrees of freedom to capture complex spectral structures.
 - High prediction error power ($E_2 \gg E_{10}$) due to incomplete energy modeling.
 - Severe loss of spectral resolution, leading to merged or missing frequency peaks.



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Root Cause Analysis

- The Burg method estimates the Power Spectral Density (PSD) by fitting an Autoregressive (AR) model $AR(P)$ to the input sequence using forward and backward prediction error minimization.
- An $AR(2)$ can model at most **one complex conjugate pole pair** (a single spectral peak/frequency resonance)
- If the underlying signal contains multiple sinusoidal components, closely spaced frequencies, or complex noise structure requiring up to 5 complex conjugate pairs ($P = 10$), the $AR(2)$ filter cannot resolve them, resulting in over-smoothed, merged, or missing spectral peaks.



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Correction

- **Impact of Model Order Selection ($P = 2$ vs $P = 10$)**
- **Poles and Spectral Resolution:**
- An AR(P) model places p poles in the z -domain. For real-valued signals, poles appear in complex conjugate pairs.
- $p = 2$: Provides only 2 poles (1 complex conjugate pair).
- It can identify at most **1 distinct spectral peak**. Any additional frequency components present in the signal are averaged together, causing severe loss of spectral resolution (underfitting).
- $p = 10$: Provides 10 poles (5 complex conjugate pairs), allowing the model to simultaneously resolve up to **5 distinct sharp spectral peaks** or accurately model complex colored noise backgrounds.



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Application of Concepts

AR Model Power Spectral Density Representation

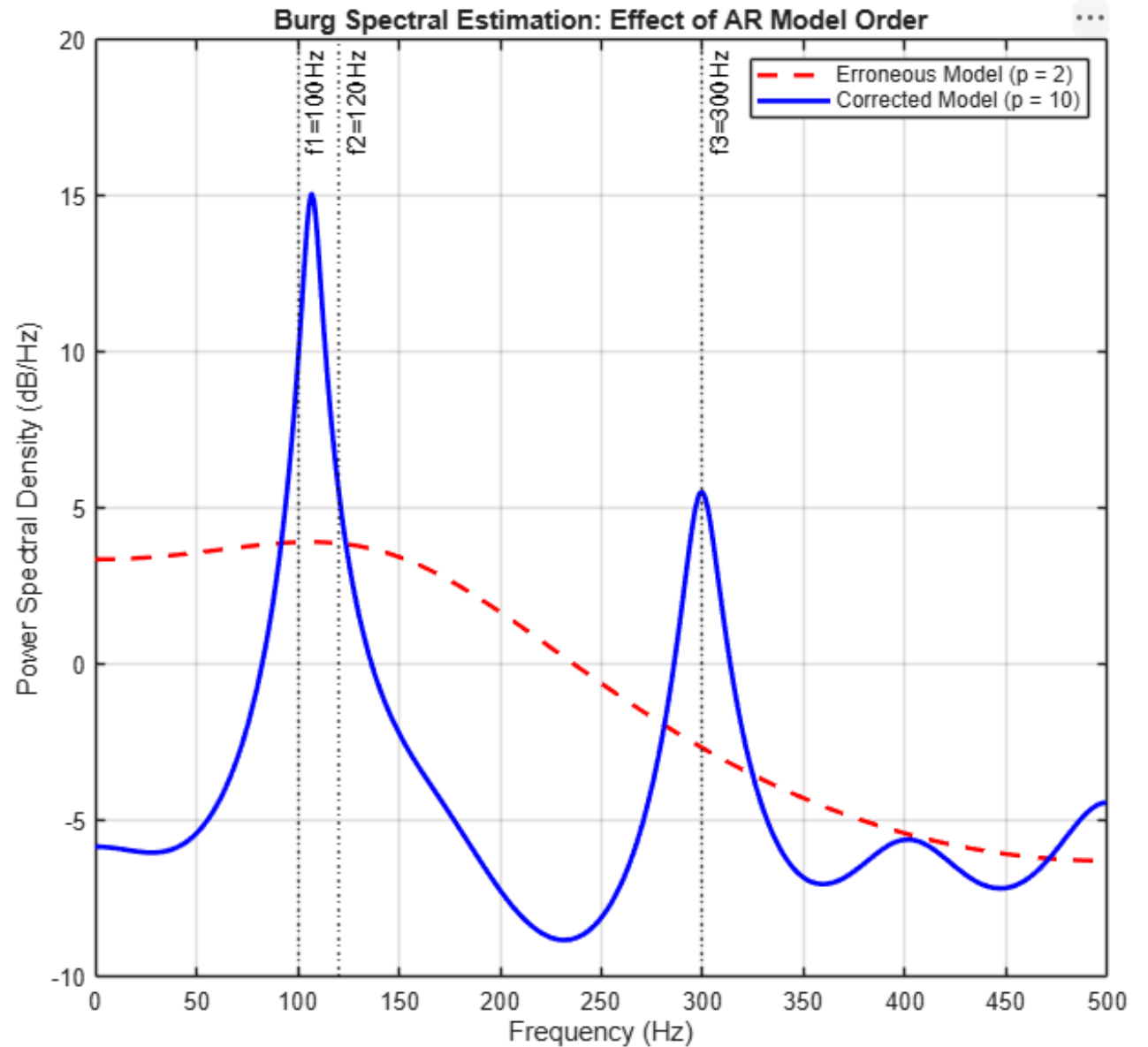
$$P_{\text{Burg}}(f) = \frac{E_p}{f_s \left| 1 + \sum_{k=1}^p a_p(k) e^{-j2\pi k f / f_s} \right|^2}$$

Where:

- p is the model order.
- E_p is the prediction error power at order p .
- $a_p(k)$ are the AR filter coefficients derived from the reflection coefficients k_p .
- f_s is the sampling frequency.

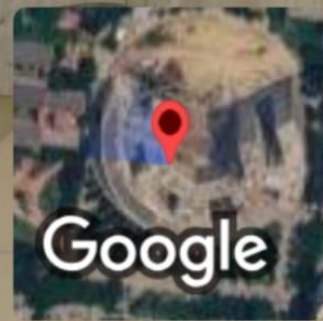
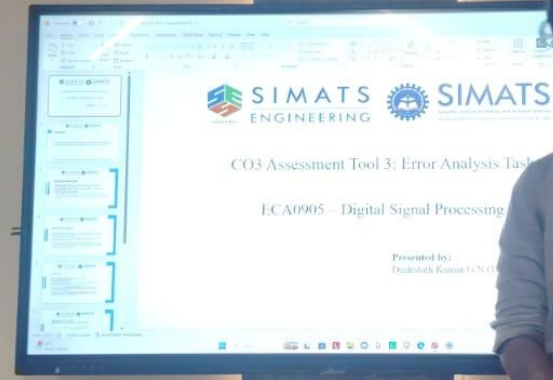
/MATLAB Drive/Dsp_CO3_AT3.m

```
1 clear; clc; close all;
2 %% 1. Generate Synthetic Signal with Multiple Spectral Peaks
3 fs = 1000; % Sampling frequency (Hz)
4 t = 0:1/fs:1-1/fs; % 1-second time vector
5 N = length(t);
6 % Signal frequencies: 100 Hz, 120 Hz, and 300 Hz
7 f1 = 100; f2 = 120; f3 = 300;
8 x = sin(2*pi*f1*t) + 0.8*sin(2*pi*f2*t) + 0.5*sin(2*pi*f3*t) + 0.5*randn(size(t));
9 %% 2. Incorrect Implementation (Model Order p = 2)
10 p_err = 2;
11 [a_err, e_err] = arburg(x, p_err); % Burg algorithm implementation
12 [h_err, w] = freqz(sqrt(e_err), a_err, 1024, fs);
13 psd_err_db = 10*log10(abs(h_err).^2);
14 %% 3. Corrected Implementation (Model Order p = 10)
15 % An AR(10) model forms 5 complex conjugate pole pairs (up to 5 peaks).
16 p_corr = 10;
17 [a_corr, e_corr] = arburg(x, p_corr); % Burg algorithm implementation
18 [h_corr, ~] = freqz(sqrt(e_corr), a_corr, 1024, fs);
19 psd_corr_db = 10*log10(abs(h_corr).^2);
20 %% 4. Plot & Compare PSD Outputs
21 figure('Color', 'w');
22 plot(w, psd_err_db, 'r--', 'LineWidth', 1.8, 'DisplayName', 'Erroneous Model (p = 2)');
23 hold on;
24 plot(w, psd_corr_db, 'b-', 'LineWidth', 2, 'DisplayName', 'Corrected Model (p = 10)');
25 xline([f1 f2 f3], 'k:', {'f1=100Hz', 'f2=120Hz', 'f3=300Hz'}, 'HandleVisibility', 'off');
26 title('Burg Spectral Estimation: Effect of AR Model Order');
27 xlabel('Frequency (Hz)');
28 ylabel('Power Spectral Density (dB/Hz)');
29 grid on;
30 legend('Location', 'northeast');
31 xlim([0 500]);
32 %% 5. Display Prediction Error Reduction
33 fprintf('=== Error Analysis Metrics ===\n');
34 fprintf('Prediction Error Power (p = 2) : %.4f\n', e_err);
35 fprintf('Prediction Error Power (p = 10): %.4f\n', e_corr);
```





Thank you !



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